Singha Junchan

Cloud-Native Security · Authorization and Access Control · Verifiable Audit · AI-Agent Security

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Intro

I'm a final-year computer engineering student at SIIT and a national programming champion (1st in Thailand, TOI18). My research starts from real problems: how to keep cloud, IoT and AI systems secure without slowing them down, and how to prove afterwards that the rules were followed. It covers zero-trust authorization for Kubernetes and multi-cloud systems, policy compilation, verifiable audit with zero-knowledge proofs, post-quantum authorization, federated threat hunting and AI-agent security. I study the problem and the published work first, then build working prototypes on Kubernetes and AWS and measure them against published baselines. First author of ten of the eleven papers below and co-author of a Best Paper. Adaptable and outgoing: tech lead of a university app, student committee president and an olympiad coach with 500+ hours of teaching. I'm seeking funded graduate study or a research internship where security meets AI, cloud, identity and governance: research internships from January 2027, graduate study after May 2027.

Education

B.Eng. in Computer Engineering Aug 2023 – May 2027 (expected)

Sirindhorn International Institute of Technology (SIIT), Thammasat University *GPA 3.67 / 4.00*

- Minor in Cloud Computing & Cybersecurity; first-class honours expected.
- Full scholarship: 4-Year Continuing Academic Outstanding Student Program.
- Research supervisor: Prof. Somchart Fugkeaw (SIIT); senior project: ZKAudit, ZK-ALBA and PQZ-Audit.

Publications

2 accepted (one Best Paper Award) · 3 submitted · 6 in preparation · first author of all but U-HAP.

Accepted

U-HAP: Unified Authorization over Heterogeneous Policies with Efficient Multi-Resource Verification in Kubernetes **Best Paper Award** · IEEE JCSSE 2026

K. Jairak, **S. Junchan**, P. Pruksorranan, S. Fugkeaw

23rd International Joint Conference on Computer Science and Software Engineering, Bangkok, June 2026; presented by S. Junchan. ACL, RBAC, ABAC and deny policies are compiled into indexes for a Kubernetes webhook authorizer: policy updates 1.6–2.7× faster than Open Policy Agent with 100/100 identical decisions, and up to 16.9× faster ABAC evaluation at 320 rules. Pages 79–84, DOI 10.1109/JCSSE68839.2026.11597088.

ZKAudit: Scalable Zero-Knowledge Auditor Authentication and Fast Integrity Verification for Cloud Logs Accepted · KST 2027

S. Junchan, S. Fugkeaw

An auditor proves credentials, scope and policy freshness without revealing them; replay-safe, load-aware verifier shards and hierarchical Merkle log commitments (Rust, AWS EC2). With batching: 1,415 proofs/s per core, 2.31× zk-promises.

Submitted

FEDAC: A Federated Dynamic Access Control Framework for Cloud-Native Systems with Privacy-Preserving Authorization and Verifiable Governance Submitted · IEEE TDSC 2026

S. Junchan, K. Jairak, P. Pruksorranan, S. Fugkeaw

A control hub verifies each caller once and issues a short-lived, service-scoped capability; a sidecar beside every workload enforces it locally, so no network call sits on the request path. Four risk-adaptive authentication tiers up to a Groth16 zero-knowledge proof; blockchain-anchored audit. On AWS EKS against three re-implemented frameworks (Heimdall, ZKP-CapBAC, Atalay): 1.4 ms median latency, about 7,100 requests/s.

PQ-SCVA-MCK: Scope-Complete Verifiable Authorization and Impact-Bounded State Continuity for Post-Quantum Secure Multi-Cluster Kubernetes Submitted

S. Junchan, S. Fugkeaw

ML-DSA-signed proofs, anchored on Hyperledger Fabric, show that every deny rule was considered. About 28,000 authorizations/s on AWS EKS, 14.5–81.6× the throughput of three earlier schemes.

SWARM-Fog: Adaptive Swarm Intelligence for Real-Time Load Sharing and Fault Recovery in Large-Scale IIoT Fog-Cloud Systems Submitted

S. Junchan, S. Fugkeaw

Self-healing industrial IoT with no central controller; fog clusters schedule by trust. 100% recovery completeness at 5–50% node failures, about 3–6× faster recovery than completeness-matched baselines.

In preparation (first author, with Prof. S. Fugkeaw)

- **ZK-ALBA**: Authorization-Bound Zero-Knowledge and Load-Sharded Auditing for Cloud Logs (journal): integrity-check cost up to 12× lower and preparation cost 13–112× lower than prior schemes.
- **PQZ-Audit**: Post-Quantum Privacy-Preserving Cloud Audit Trails with Verifiable Predicate Execution and Threshold Authorization. Auditors run real queries over encrypted logs (time windows, thresholds, totals) and verify the results are correct and complete without decrypting them; t-of-n post-quantum approval for privileged audits.
- **FRESCO**: Freshness-Aware Elastic Orchestration of Stateful Services for the Multi-Cloud Continuum. Sparse-Merkle proofs keep scheduling and failover off stale replicas: 0% violations on AWS EKS (baselines: 65–67%).
- **MoCA** (journal): containment of AI agents after a successful prompt injection: a compromised agent cannot exceed its signed human delegation, checked at an MCP gateway and at the resource.
- **Threshold-gated encrypted SIEM**: a SIEM that works on encrypted logs shared by several organisations; tracing across them opens only when enough of them report the same signal.
- **Fed-RAG-Hunt**: Differentially-Private Latent Sharing and Open-Set Prototype Classification for Federated Zero-Day Threat Hunting. Members share private fingerprints of suspicious activity, not data, so a threat seen by one alerts all.

Experience

Tech Lead – *TU Alert, built for Thammasat University, launching soon* 2026 – Present

- Lead a two-student team hired by the university to build a LINE (LIFF) incident-reporting system that routes each report to the responsible office; design the architecture and security model, and contribute code (TypeScript, Next.js).
- Security by design: two-factor staff sign-in, data retention and erasure, Semgrep and gitleaks in CI.

Information Security Intern – *Allianz Ayudhya, Bangkok* Jun – Jul 2026

- Built two AI agents for third-party risk management that support human judgement: a questionnaire agent that drafts due-diligence answers with evidence, a reason and a confidence score, identifies gaps and recommends fixes, all human-reviewed; and a guidance agent that coaches business owners step by step. Tested on real, anonymised cases.
- Learned governance and third-party risk in practice (ISO 27001, NIST, internal policies), mentored by the CISO.

R&D Intern, AI Research Team – *Amity Solutions, Bangkok* Jul – Aug 2023

- Translated new LLM research into working prototypes and assessed the latest GPT and Claude models; built knowledge-base chatbots with retrieval-augmented generation (RAG), tested end to end.

Teaching and Mentoring

Founder & Lead Instructor – *TOI Tutor and ABIG Center, online* 2022 – Present

- Founded two olympiad-coaching platforms (TOI, POSN): 500+ hours of teaching, hundreds of students; wrote COM300, C++ and algorithms (about 270 pages, 128 recorded lessons), and COM100, mathematics and Python.

Honours and Awards

- **Best Paper Award**, IEEE JCSSE 2026, for U-HAP · **1st in Thailand**, national champion and gold medal, Thailand Olympiad in Informatics (TOI18), 2022, after more than 1,000 hours of training.
- 1st place (team), Image Recognition Programming Challenge, Osaka Institute of Technology and SIIT, 2025.
- Full scholarship, 4-Year Continuing Academic Outstanding Student Program, SIIT, 2023 – 2027.
- Bronze Medal & Popular Vote, SCiUS Forum 2022 (IoT cat feeder) · winner, HotelDirect.in.th startup, depa Thailand Digital BIG BANG Regional, 2018 · bronze medal, National Robot Competition, 2017.

Certifications and Security Training

- ISC2 Certified in Cybersecurity (CC), 2025 · CompTIA Security+ (exam planned Dec 2026).
- SOSECURE Road to Cybersecurity 2025 (GEN6): eight modules, from networking and Linux hardening to web pentesting and SOC (Suricata, Splunk, Wazuh) · DataCamp: Azure and AWS, 2025.
- NCSA CTF Thailand Tournament 2026, as a team · home lab on Kali Linux · TryHackMe, Hack The Box.

Skills

Cryptography:	zk-SNARKs (Groth16), Merkle commitments, post-quantum ML-DSA and ML-KEM
Cloud & Systems:	Kubernetes (RBAC, ABAC), AWS (EKS, EC2), Docker, Hyperledger Fabric, OPA/Rego
Security Tools:	Nmap, Wireshark, Burp Suite, Metasploit, Splunk and Wazuh (SOC labs), OWASP Top 10
AI:	LLM agents, RAG, MCP, prompt-injection testing
Programming:	Python and C++ (main); also Go, Rust, TypeScript, C, Java, Bash, SQL
Languages:	Thai (native), English (fluent), Chinese (basic)
Interests:	home-lab security research, Muay Thai, board games, food and travel

Leadership and Service

- President, Sub Student Committee 32, SIIT (2024 – present): leads a committee that represents the student body; organised 11 student events and led SIIT First Meet 2024 (about 500 participants, 100 staff).
- President, Psychology Club, Darunsikkhalai Science School, 2021.

Referees

Prof. Somchart Fugkeaw – research supervisor; co-author of all the papers above
School of ICT, SIIT, Thammasat University *somchart@siit.tu.ac.th*

Apichai Wongchindakhun – internship mentor
Chief Information Security Officer (CISO), Allianz Ayudhya *linkedin.com/in/apichai-wongchindakhun-1839b314*